GW signal buried in strain data are modeled in terms of the parameters $\vec{\Theta}$ described in section II A as $h(t;\Theta)$. Under these assumptions, search pipelines use matched filter techniques to estimate the SNR as,

$$\rho^2 = 4\int_{f_{\rm low}}^{f_{\rm high}} \frac{|\tilde{s}(f)\tilde{h}(f,\Theta)|}{S_n(f)} df, \tag{3}$$

where $\tilde{s}(f)$ and $\tilde{h}(f)$ are the functions of frequency and are the Fourier transforms of the time domain quantities $s(t)$ and $h(t)$, respectively. $S_n(f)$ is the power spectral density, which quantifies the noise properties of the GW detector. $f_{\rm low}$ and $f_{\rm high}$ are the low and high-frequency cutoffs for the integration limit. A template bank-based approach is employed in data analysis pipelines for GW searches [85–87]. The SNR time series is evaluated for each point in the template bank. When the SNR surpasses a certain threshold, it is deemed a trigger and is saved for further analysis. Triggers from individual detectors that occur within the light travel time between the detectors are known as coincident triggers. In the next stage of GW searches, these coincident triggers are further examined, and their significance is estimated by comparing them with the expected distribution of background triggers. Statistically significant triggers are classified as GW merger events [84, 86]. A more detailed analysis is done using PE techniques to determine the waveform model's parameters ($\vec{\Theta}$) [88, 89].

In order to estimate the parameters of a GW merger signal, the standard method is to use a Bayesian framework. For the given data d, the likelihood function $\mathcal{L}(d|\vec{\Theta}, I)$ describes the probability of obtaining the data for a given model with any other prior information I. The posterior probability distribution $p(\vec{\Theta}|d, I)$ is then calculated using Bayes' theorem,

$$P(\vec{\Theta}|d, I) = \frac{\mathcal{L}(d|\vec{\Theta}, I)\pi(\vec{\Theta}|I)}{p(d|I)} \tag{4}$$

where $\pi(\vec{\Theta}|I)$ represents the prior probability distribution for parameters $\vec{\Theta}$ and $p(d|I)$ describes the marginalised likelihood.

C. Sources of errors in the parameter estimation

There can be systematic errors present in the PE analysis due to any of the following reasons:

- **Data analysis artifacts due to mis-modeling the noise:** For certain periods of strain data, the noise in the GW detector may not adhere to the assumptions of Gaussianity and stationarity. This deviation can occur due to environmental factors or issues with the instrument itself. In a study, Mozzon et al. [90] found that approximately three percent of the GW detector data does not conform to the assumptions of Gaussianity and stationarity. This represents a situation where the noise model fails. If left unaddressed, this could result in biased estimates of the properties of the binary source (systematic errors) or inaccurate statistical uncertainties (statistical errors) [29, 91].

- **Waveform systematics:** With the improvements in the sensitivity of the GW detectors and with the addition of more physics in the description of waveform from the GW merger, such as the inclusion of higher modes, eccentricity, precision, the requirements for the accuracies of the waveform models are becoming stricter. We have already reached an era where the systematics of the waveform models can not be ignored. If the waveform model does not describe the underlying reality, and if we do not account for the possible errors, the estimates of source properties are expected to be biased.

- **Other unaccounted effects:** Even if our models for noise and the waveform describe the strain data perfectly, there can be rare cases such as modification of GW waveform by strong lensing by intermediate-mass black hole object or presence of sub-threshold overlapping signals, etc. It can give rise to unmodeled effects, which can again bias the PE results.

It is crucial to consider any systematic errors that may arise from the above mentioned effects. This work focuses on the systematic and statistical errors arising from inaccuracies in waveform models. In PE, the systematic errors can be represented by $|\Delta\mathbf{\Theta}| = |\mathbf{\Theta}_{\rm rec} - \mathbf{\Theta}_{\rm true}|$, where $\mathbf{\Theta}_{\rm true}$ represents the true value of waveform parameters and $\mathbf{\Theta}_{\rm rec}$ represents the recovered value of the parameters. For individual waveform parameters, we define $\Delta\Theta^i = |\Theta^i_{rec} - \Theta^i_{true}|$ where Θ^i_{rec} is the median value of one dimensional marginalized posterior samples for i-th waveform parameter. Similarly, the statistical error σ_i represents the standard deviation of marginalized posterior distribution of i-th waveform parameter. For a given waveform model, the statistical errors depend on the signal loudness and detector sensitivity. We can ignore systematic errors when,

$$|\Delta\Theta^i| << \sigma_i \quad \forall \quad i. \tag{5}$$

Conversely, the systematic errors become important when,

$$\frac{|\Delta\Theta^i|}{\sigma_i} \sim \mathcal{O}(1) \quad \text{for any i} \tag{6}$$

III. MODELING UNCERTAINTIES IN STRAIN

The description of transient signal $h(t, \vec{\Theta})$ can be affected by following:

- **Calibration uncertainties:** The GW detectors are calibrated periodically to map between input

strain and time series output. GW detectors are not perfectly calibrated and the calibration uncertainties are provided with detector characterization [92].

- **Inaccuracies in waveform modeling:** These inaccuracies might arise due to one of many reasons, such as i) errors in calibrating to NR waveforms, ii) not accounting for all the physics in the waveform modeling, iii) inherent errors in NR simulations.

To model general uncertainties in the observed strain data, we go to the frequency domain where the signal, $\tilde{h}_{\rm obs}(f)$, can be modeled as,

$$\tilde{h}_{\rm obs}(f) = \tilde{A}_{\rm obs}(f)\exp\{\iota\tilde{\phi}_{\rm obs}(f)\}, \tag{7}$$

where $A_{\rm obs}(f)$ and $\phi_{\rm obs}(f)$ are observed amplitude and phase of the signal. From now on, we will drop writing the explicit frequency dependence. Whenever we present a quantity in the frequency domain, with overhead $\tilde{}$, it is assumed that it is generally an explicit function of frequency unless stated otherwise. Now we assume that the observed signal $\tilde{h}_{\rm obs}$ can be expressed in terms of 'true' amplitude and phase with linear perturbation,

$$\begin{aligned}\tilde{h}_{\rm obs} &= \tilde{A}_{\rm obs}\exp\{\iota\tilde{\phi}_{\rm obs}\} \\ &= (\tilde{A}_{\rm true}+\delta\tilde{A}_{\rm abs})\exp\{\iota(\tilde{\phi}_{\rm true}+\delta\tilde{\phi}_{\rm abs})\} \quad (8)\\ &= \tilde{A}_{\rm true}(1+\delta\tilde{A}_{\rm rel})\exp\{\iota\tilde{\phi}_{\rm true}(1+\delta\tilde{\phi}_{\rm rel})\} \quad (9)\end{aligned}$$

where $\tilde{A}_{\rm true}$ and $\tilde{\phi}_{\rm true}$ are true amplitude and phase of the signal. $\delta\tilde{A}_{\rm abs}$ and $\delta\tilde{\phi}_{\rm abs}$ are absolute errors in the true amplitude and phase, respectively, while $\delta\tilde{A}_{\rm rel}$ and $\delta\tilde{\phi}_{\rm rel}$ are relative deviation in true amplitude and phase respectively. The relation between absolute and relative deviation (or errors) are, $\delta\tilde{A}_{\rm rel} = \frac{\delta\tilde{A}_{\rm abs}}{\tilde{A}_{\rm true}}$ and $\delta\tilde{\phi}_{\rm rel} = \frac{\delta\tilde{\phi}_{\rm abs}}{\tilde{\phi}_{\rm true}}$.

A. Detector calibration uncertainties

When GW passes through the detector, it produces a differential arm displacement between two L-shaped arms relative to the original arm lengths. This differential arm displacement generates laser power fluctuations in the photodetector, reproduced as the time series data. The conversion to the digital strain data regarding time series from the differential arm displacement happens through the control loop and calibration pipeline [93, 94]. The calibration of the strain data are modeled with systematic and statistical errors, which are then provided as calibration uncertainties as function of frequency [93–97]. Figure 1 shows the calibration envelopes for the LIGO-Hanford detector around the event GW150914.

The calibration uncertainties are modeled as frequency dependent errors in phase and amplitude as follows [98, 99]:

$$\tilde{h}_{\rm obs} = \tilde{h}_{\rm true}(1+\delta\tilde{A}_{\rm cal})\exp(\iota\delta\tilde{\phi}_{\rm cal}), \tag{10}$$

In terms of the general parametrization we discussed above, $\delta\tilde{A}_{\rm cal}$ can be identified as relative error in amplitude while $\delta\tilde{\phi}_{\rm cal}$ is absolute error in the phase. In the calibration uncertainty model, equation (10), the exponential term can be expanded as a Taylor series and is rewritten as,

$$\tilde{h}_{\rm obs} = \tilde{h}_{\rm true}(1+\delta\tilde{A}_{\rm cal})\frac{2+\iota\delta\tilde{\psi}_{\rm cal}}{2-\iota\delta\tilde{\psi}_{\rm cal}}, \tag{11}$$

where the ratio involving $\delta\tilde{\psi}$ is designed to have a modulus of unity, thus being a pure phase [98]. In existing PE pipelines such as PyCBC-Inference [89] and BilBy [100], the projected strain in each detector is modeled with phase and amplitude correction using cubic-spline curves drawn from the distribution provided by calibration envelopes, as shown in Figure 1.

B. GW waveform model uncertainties

Motivated by how detector calibration uncertainties are handled, we suggest that uncertainties in waveform models can be treated similarly. Specifically, we propose parametrizing the uncertainties in the description of GW waveform in the frequency domain. Furthermore, we suggest that waveform developers provide uncertainty bands with their model. A similar approach was proposed in [32]. Here, we examine the technical details, feasibility, choices of parametrization, and degeneracy with the detector calibration-correction framework. We use the following parametrizations to account for waveform modeling errors:

- **abs-phase**: Relative errors in amplitude and absolute errors in phase:

$$\begin{aligned}\tilde{h}_{model} &= \tilde{A}_0(1+\delta\tilde{A})\exp\{\iota(\tilde{\phi}_0+\delta\tilde{\phi})\}, \\ &= \tilde{h}_0(1+\delta\tilde{A})\exp\{\iota\delta\tilde{\phi}\},\end{aligned} \tag{12}$$

- **rel-phase**: Relative errors in amplitude and phase:

$$\begin{aligned}\tilde{h}_{model} &= \tilde{A}_0(1+\delta\tilde{A})\exp\{\iota\tilde{\phi}_0(1+\delta\tilde{\phi}_{\rm rel})\}, \\ &= \tilde{h}_0(1+\delta\tilde{A})\exp\{\iota\tilde{\phi}_0\delta\tilde{\phi}_{\rm rel}\},\end{aligned} \tag{13}$$

where $\tilde{h}_0 = \tilde{A}_0\exp(\iota\tilde{\phi}_0)$ represent the baseline (or reference) model, which aspires to be a 'true' model. As no waveform model is perfect to arbitrary accuracy, we introduce additional terms to account for potential systematics. $\delta\tilde{A}$ represent the fractional or relative error in the amplitude of the signal, and $\delta\tilde{\phi}$ ($\delta\tilde{\phi}_{\rm rel}$) represent absolute (relative/fractional) errors in phase.

The abs-phase parametrization described by Eq. (12) is also used in the correction for the detector calibration. Other studies also use this parametrization to quantify the waveform uncertainties [32, 101]. Since $\delta\tilde{A}$ is the relative change in amplitude, it is a dimensionless parameter in these parametrizations. The units of $\delta\tilde{\phi}$ is radian in

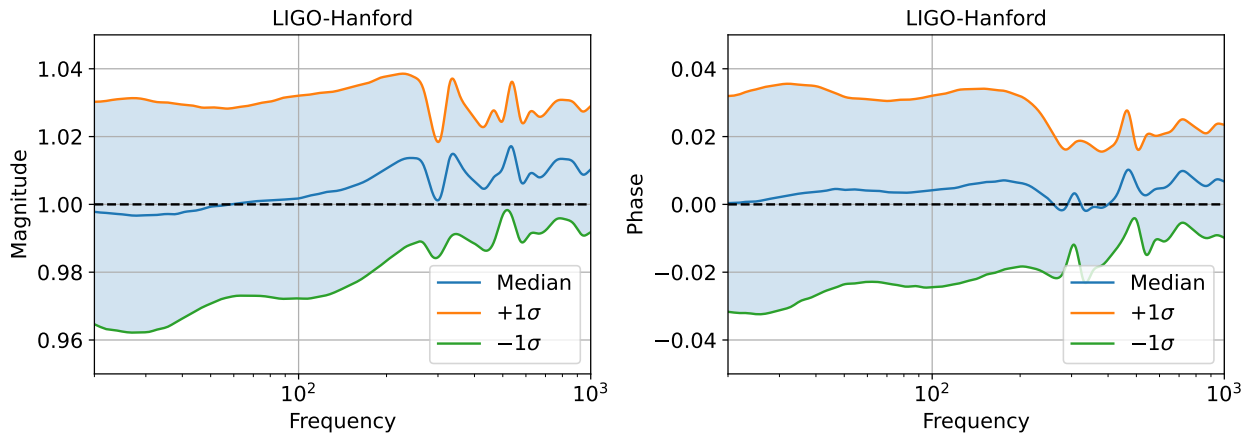

FIG. 1. The Figure shows the calibration uncertainties as a function of frequency for the LIGO Hanford detectors around the time of the first GW detection: GW150914. The left panel shows the systematic error in the magnitude, while the right panel shows the systematic error in the phase angle (in radians). The shaded region represents the $\pm 1\sigma$ band around the median systematic error. The systematic error is described by the ratio of the true response function to the modeled response function of the detector. The LIGO and Virgo calibration uncertainty files for O1, O2, and O3 observation runs are available at [92].

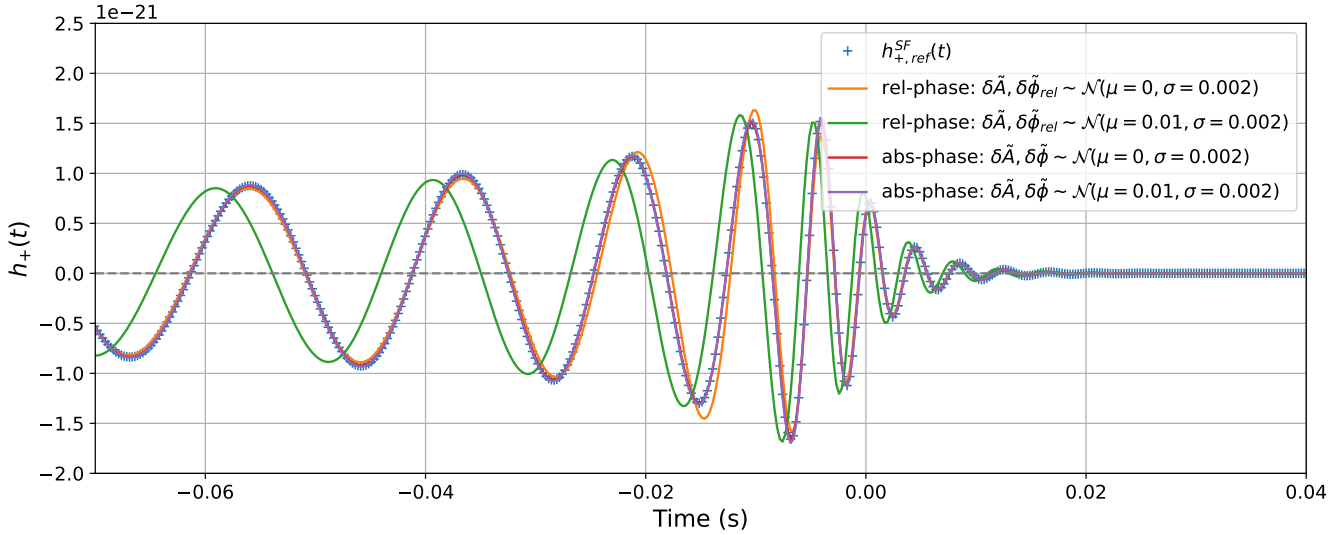

FIG. 2. We show the time domain waveform in a signal frame (before projecting it in the detector) for one of the polarization (+) for a GW150914-type signal. Apart from the reference waveform model `IMRPhenomPv2` $h^{SF}_{+,ref}(t)$ (blue, denoted by '+' markers), we also show the modification to the reference signal by one of the parametrizations described in the text. Red and violet waveforms represent the abs-phase modification given by (12). We use cubic splines for the parameters $(\delta\tilde{A}, \delta\tilde{\phi})$ with a realization from the normal distribution as shown in corresponding plot legends. Orange and green waveforms represent the modification with rel-phase parametrization given by equation (13).

abs-phase parametrization while $\delta\tilde{\phi}_{\rm rel}$ is a dimensionless parameter.

A point of caution must be mentioned here for description of the relative phase error. The baseline phase $\tilde{\phi}_0$ is only uniquely defined up to integer multiples of 2π. However, the multiplication with $\delta\tilde{\phi}_{\rm rel}$ would give the phase error a different physical meaning depending on the chosen branch of the baseline phase. Therefore, a convention must be to introduced to remove this ambiguity. We implement a simple method that ensures the baseline phase to be between $-\pi$ and π at the starting frequency of the waveform generation. The rest of the phase evolution follows a standard unwrapping algorithm as implemented in Numpy [102].

We note that alternative choices for removing this ambiguity exist. For example, one could parameterize the phase by an exact value $\tilde{\phi}_{\rm ref}$ at a given reference frequency $f_{\rm ref}$. By this convention, the error would vanish